Jichao (Ciao) Yu

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EDUCATION

University of Georgia 2024 - Present

Ph.D. student in Computer Science; expected May 2029

Advisor: Ninghao Liu

The Ohio State University May 2024

M.S. in Electrical & Computer Engineering

Advisor: Jia Liu | GPA: 3.66/4.0

Wuhan University of Science and Technology June 2021

B.Eng. in Electronic Information Engineering

GPA: 3.76/4.0 | Outstanding Graduate Thesis Award

RESEARCH INTERESTS

Machine learning systems, federated and distributed learning, optimization, and AI for education.

RECENT PUBLICATIONS AND PREPRINTS

[1] Zhongzhi Li, Yucheng Shi, Zongxia Li, Junyao Yang, Ruhan Wang, Yu Wang, Jingyuan Huang, **Jichao Yu**, Ninghao Liu, Haitao Mi, Leowei Liang. **Self-Supervised Scaling of Terminal Environments for Scientific Domains**. arXiv preprint, 2026. arXiv:2610.02710.

[2] Shuchen Guo*, Yun Wang*, **Jichao Yu***, Xuansheng Wu*, Bilgehan Ayik, Field M. Watts, Ehsan Latif, Ninghao Liu, Lei Liu, Xiaoming Zhai. **Artificial Intelligence Bias on English Language Learners in Automatic Scoring**. Artificial Intelligence in Education (AIED 2025), LNCS 15881, pp. 268-275, 2025. doi:10.1007/978-3-031-98462-4_34.

** Equal contribution (co-first authors).*

[3] Ehsan Latif, Yifan Zhou, Shuchen Guo, ..., **Jichao Yu**, ..., Xiaoming Zhai. **A Systematic Assessment of OpenAI o1-Preview for Higher Order Thinking in Education**. arXiv preprint, 2024. arXiv:2410.21287.

[4] Tianyang Zhong, Zhengliang Liu, Yi Pan, ..., **Jichao Yu**, ..., Tianming Liu. **Evaluation of OpenAI o1: Opportunities and Challenges of AGI**. arXiv preprint, 2024; revised September 2026. arXiv:2409.18486.

EARLIER PUBLICATIONS

[5] Xiaogao Yu, **Jichao Yu**. **Analysis on the technical framework of virtual college students' mental health service center**. Changjiang Information & Communications, 35(12), pp. 123-126, 2022 (in Chinese).

[6] **Jichao Yu**, Xiaogao Yu. **Prediction Model Hadoop-based for High-risk Students**. MMSTA 2019, Advances in Computer Science Research 93, pp. 30-33, 2019. doi:10.2991/mmsta-19.2019.7.

[7] **Research on Personalized Learning Space in Internet and Big Data Environment**. Software Guide, issue 12, 2019 (in Chinese). Author: **Jichao Yu**.

[8] Xiaogao Yu, **Jichao Yu**. **Features Selection of Exponential Distribution-based Unbalanced Data**. Statistics & Decision, issue 20, 2019 (in Chinese).

[9] **Research on Prediction of 'Four Challenged Students' Based on Big Data**. Information and Communications, issue 9, 2019 (in Chinese). Co-author: **Jichao Yu**.

RESEARCH EXPERIENCE

Federated Learning Started May 2023
The Ohio State University, Department of Electrical and Computer Engineering
Advisors: Jia (Kevin) Liu and Haibo Yang

- Trained heterogeneous federated learning models with PyTorch using ResNet and Transformer architectures on CIFAR-10, CIFAR-100, and WikiText.

Space Satellite Detection Dec. 2022 - May 2023
The Ohio State University, Department of Electrical and Computer Engineering
Advisor: Wladimiro Villarroel

- Preprocessed satellite images with MATLAB and trained a ResNet model in Python; achieved 99% classification accuracy.

Prediction and Evaluation of Teaching Effectiveness Jan. 2023 - May 2023
Hubei University of Economics
Advisor: Xiaogao Yu

- Optimized a weighted voting ensemble in Python for a Chinese Society of Higher Education digital course resources project (21SZYB01); achieved 99.3% accuracy.

Mental Health Risk Warning and Crisis Intervention Nov. 2021 - Dec. 2022
Hubei University of Economics | Research Assistant
Advisor: Xiaogao Yu

- Extracted learning psychology and mental health features in Python for a Hubei Province philosophy and social sciences research project (21ZD092).